int, str, bool, float

LIST, TUPLE, DICTIONARY, SET

In [1]:
```
type(True)
```

Out[1]: bool

In [2]:
```
# Keywords and Variables
True
int
float
tuple
```

Out[2]: tuple

In [5]:
```
num = 5.567
```

1. Letter, Underscore - start
2. No empty spaces
3. Keywords cannot be used as variables

In [11]:
```
# List
l = list()
print(l)

m = []
print(m)
```

```
[]
[]
```

In [13]:
```
a = ['Shreyas', 'Druthi', 'Ambika']
print(a)
type(a)
```

```
['Shreyas', 'Druthi', 'Ambika']
```

Out[13]: list

In [14]:
```
b = [1, 2.2, True, 'Shreyas']
print(b)
type(b)
```

```
[1, 2.2, True, 'Shreyas']
```

Out[14]: list

In [15]:
```
# List Indexing - Positive Indexing and Negative Indexing

t = [1, 2.2, True, 'Shreyas']
#    0   1    2      3
```

```
t[3]
```

Out[15]: 'Shreyas'

In [16]:
```
t = [1, 2.2, True, 'Shreyas']
#   -4   -3      -2       -1

t[-1]
```

Out[16]: 'Shreyas'

In [17]:
```
len(t)
```

Out[17]: 4

In [19]:
```
l = [1, [2,3], 4]
print(l[0],l[1],l[2])
```

```
1 [2, 3] 4
```

In [21]:
```
l[1][1]
```

Out[21]: 3

In [22]:
```
m = [      [1,2,3]      ,     'Nice'       , [7]         ]
```

In [23]:
```
len(m)
```

Out[23]: 3

In [24]:
```
m[-1]
```

Out[24]: [7]

In [25]:
```
m[0][1]
```

Out[25]: 2

Slicing

In [30]:
```
l = list(range(2, 11, 2))
l
```

Out[30]: [2, 4, 6, 8, 10]

In [31]:
```
n = list(range(1,11))
n
```

Out[31]: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

In [32]:
```python
p1 = n[0:5]
p1
```
Out[32]: [1, 2, 3, 4, 5]

In [34]:
```python
p2 = n[4:10]
p2
```
Out[34]: [5, 6, 7, 8, 9, 10]

In [35]:
```python
p3 = n[:5]
p3
```
Out[35]: [1, 2, 3, 4, 5]

In [36]:
```python
[1,2,3,4,5,6,7,8,9,10]
#       -6  -5   -4  -3  -2  -1
```
Out[36]: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

In [37]:
```python
p4 = n[4:]
p4
```
Out[37]: [5, 6, 7, 8, 9, 10]

In [38]:
```python
p5 = n[:]
p5
```
Out[38]: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

In [44]:
```python
n
```
Out[44]: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

In [47]:
```python
p6 = n[::2]
p6
```
Out[47]: [1, 3, 5, 7, 9]

List Comprehension

In [48]:
```python
a = list(range(1,11))
a
```
Out[48]: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

In [49]:
```python
b = [x for x in range(1,11)]
b
```

Out[49]: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

In [50]:
```
c = [(x/2) for x in range(1,11)]
c
```

Out[50]: [0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0]

In [51]:
```
d = [(x**2) for x in range(1,11)]
d
```

Out[51]: [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]

List Concatenation

In [52]:
```
a = list(range(1,6))
b = list(range(6,11))
print(a,b)
```

```
[1, 2, 3, 4, 5] [6, 7, 8, 9, 10]
```

In [53]:
```
c = a + b
print(c)
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

In [54]:
```
a = list(range(1,6))
b = list(range(4,11))
print(a,b)
```

```
[1, 2, 3, 4, 5] [4, 5, 6, 7, 8, 9, 10]
```

In [57]:
```
d = set(a) - set(b)
print(list(d))
```

```
[1, 2, 3]
```

Tuples

In [58]:
```
a = tuple()
a
```

Out[58]: ()

In [59]:
```
b = ()
b
```

Out[59]: ()

In [61]:
```
type(b)
```

Out[61]: tuple

In [62]:
```python
# List is Mutable, Tuple is Immutable
a = [1,2,3]
a[-1] = 4
print(a)
```

```
[1, 2, 4]
```

In [66]:
```python
b = (1,2,3)
b
```

Out[66]:
```
(1, 2, 3)
```

Dictionary

In [67]:
```python
d = dict()
d
```

Out[67]:
```
{}
```

In [68]:
```python
e = {}
e
```

Out[68]:
```
{}
```

In [69]:
```python
f = {'Shreyas' : '3rd Year', 'Druthi' : '2nd Year', 'Ambika' : 'Graduated'}
f
```

Out[69]:
```
{'Shreyas': '3rd Year', 'Druthi': '2nd Year', 'Ambika': 'Graduated'}
```

In [70]:
```python
f['Druthi']
```

Out[70]:
```
'2nd Year'
```

In [71]:
```python
f.keys()
```

Out[71]:
```
dict_keys(['Shreyas', 'Druthi', 'Ambika'])
```

In [72]:
```python
f.values()
```

Out[72]:
```
dict_values(['3rd Year', '2nd Year', 'Graduated'])
```

In [76]:
```python
f = {'Shreyas' : '3rd Year', 'Druthi' : '2nd Year', 'Ambika' : 'Graduated', 'S
f
```

Out[76]:
```
{'Shreyas': '4th Year', 'Druthi': '2nd Year', 'Ambika': 'Graduated'}
```

In [ ]:
```python
d = {101 : 'A', 102 : 'B'}
```

In [78]:
```python
a=5
b=7
```

```
result=a+b
print("sum:",result)
```

```
sum: 12
```

In [82]:

```
length=10
width=4
area=length*width
print("area of rectangel:",area)
```

```
area of rectangel: 40
```

In [84]:

```
principal=1000
rate=5
time=2
simple-intrest=(principal*rate*time)/100
print("simple intrest:",simple-intrest)
```

```
  Cell In[84], line 4
    simple-intrest=(principal*rate*time)/100
    ^
SyntaxError: cannot assign to expression here. Maybe you meant '==' instead of
'='?
```

In [85]:

```
a=10
b=20
a,b=b,a
print("after swapping:a=","b=",b)
```

```
after swapping:a= b= 10
```

In [87]:

```
principal=100
rate=5
time=2
simple-intreast=[principal*rate*time]/100
print("simple intrest:",simple-intrest)
```

```
  Cell In[87], line 4
    simple-intreast=[principal*rate*time]/100
    ^
SyntaxError: cannot assign to expression here. Maybe you meant '==' instead of
'='?
```

num=4 square=num**2 cube**3 print("Square:",square) print("Cube:",cube)

In [89]:

```
a=10
b=20
c=30
average=(a+b+c)/3
print("Average:",average)
```

```
Average: 20.0
```

In [ ]: